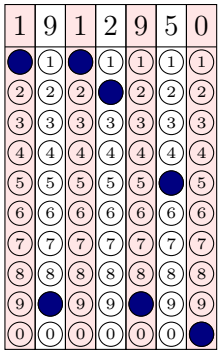
TEST VARIANTI RAQAMI: **1912950**

# MATEMATIKA

Ushbu test varianti 30 ta test topshirig‘idan iborat. Uni bajarishga 1 soat vaqt belgilash tavsiya etiladi. Natijani bilish uchun javoblaringizni javoblar varaqasiga belgilab, rasmini **@dtmxizmatbot** orqali jo‘nating. Sizga omad tilaymiz!

TEST VARIANTI RAQAMINI JAVOBLAR VARAQASIGA TO‘G‘RI KO‘CHIRING!




**1.** $A = \{a; b; c; d; e; f\}$ to‘plamning nechta qism to‘plamlarida, $b$ elementi bo‘lib, $c$ elementi qatnashmaydi?

A) *8* B) *28* C) *32* D) *16*

**2.** $\sqrt{5 - 11x - 3x^2} \geq 1$ tengsizlikning butun yechimlari sonini toping.

A) *5* B) *3* C) *4* D) *2*

**3.** $0,009 \cdot 0,02 \cdot 10^6$ ko‘paytmani standart shaklga keltiring.

A) $1,8 \cdot 10^{-2}$ B) $1,8 \cdot 10^{-1}$ C) $1,8 \cdot 10$ D) $1,8 \cdot 10^2$

**4.** To‘g‘ri chiziqda bir biri bilan ustma-ust tushmaydigan 5 ta nuqta tanlangan. To‘g‘ri chiziqda jami nechta kesma hosil bo‘ladi?

A) *10* B) *7* C) *9* D) *4*

**5.** Sayyoh belgilangan yo‘lning $\frac{1}{6}$ qismini bosib o‘tgach, yo‘l yarmigacha yana 16 km qolgan bo‘lsa, belgilangan yo‘l uzunligini (km) toping.

A) *36* B) *42* C) *54* D) *48*

**6.** $a$ ning qanday qiymatlarida uzunliklari mos ravishda $a + 2$; 4 va $2a - 1$ bo‘lgan kesmalardan uchburchak yasash mumkin?

A) $(0; 7)$ B) $(0; 5)$ C) $(1; 7)$ D) $(1; 8)$

**7.** $\left(\frac{1}{\sin^2\alpha} - \frac{1}{\cos^2\alpha}\right) \cdot \frac{\sin 2\alpha \operatorname{tg}\alpha}{1 - tg^2\alpha}$ ifodani soddalashtiring.

A) 2 B) $\sin\alpha$ C) $\cos\alpha$ D) 1

**8.** $\alpha$ tekislik va uni kesib o‘tmaydigan $AB$ kesma berilgan. Kesmaning uchlaridan $\alpha$ tekislikkacha bo‘lgan masofalar $AA_1$=18 cm, $BB_1$=13 cm bo‘lsa, $AB$ kesmani $A$ uchidan boshlab hisoblaganda 3:2 nisbatda bo‘luvchi $C$ nuqtadan $\alpha$ tekislikkacha bo‘lgan masofani (cm) toping.

A) *15,5* B) *16* C) *15* D) *14*

**9.** $\frac{1}{3}$ va $\frac{5}{6}$ sonlar orasida maxraji 24 bo‘lgan, qisqarmaydigan barcha kasrlar yig‘indisini toping.

A) *2,5* B) $4\frac{1}{6}$ C) *1,5* D) *5,5*

**10.** Hisoblang:
$139 \cdot 163 - 160 \cdot 139 + 141 \cdot 175 - 172 \cdot 141$

A) *840* B) *870* C) *864* D) *852*

**11.** O‘tkir burchakli uchburchakning ikki tomonining uzunliklari ayirmasi 2 cm ga teng, bu tomonlarining uchinchi tomonga proyeksiyalari 9 cm va 5 cm bo‘lsa, uchburchakka tashqi chizilgan aylana radiusini toping.

A) $6\frac{3}{10}$ B) $7\frac{2}{7}$ C) $5\frac{5}{12}$ D) $8\frac{1}{8}$

**12.** $m$ ning qanday qiymatida $4x^2 - 5x + m = 0$ tenglamaning $x_1$ va $x_2$ ildizlari $4x_1 + 3x_2 = 3\frac{1}{2}$ tenglikni qanoatlantiradi?

A) *−1,5* B) *−6* C) *1,5* D) *6*

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**13.** $\int x \cdot \cos 3x dx$ integralni hisoblang.

A) $-\frac{x}{3} \cdot \sin 3x - \frac{1}{9}\cos 3x + C$

B) $-\frac{x}{3} \cdot \sin 3x + \frac{1}{9}\cos 3x + C$

C) $\frac{x}{3} \cdot \sin 3x - \frac{1}{9}\cos 3x + C$

D) $\frac{x}{3} \cdot \sin 3x + \frac{1}{9}\cos 3x + C$

**14.** 2 ta har xil kitobni 12 ta o‘quvchidan 2 tasiga bittadan berish sharti bilan necha xil usulda berish mumkin?

A) *132* B) *156* C) *78* D) *66*

**15.** Hisoblang: $\sqrt{3-2\sqrt{2}} + \sqrt{3+2\sqrt{2}}$

A) $\sqrt{2}-1$ B) $\sqrt{2}+1$ C) $\sqrt{2}$ D) $2\sqrt{2}$

**16.** $\frac{\sqrt{10-3x}}{\log_2|x-3|} < 0$ tengsizlikni yeching.

A) $\left(3; 3\frac{1}{3}\right)$ B) $(2;3) \cup \left(3; 3\frac{1}{3}\right)$

C) $\left(3; 3\frac{1}{3}\right) \cup \left(3\frac{1}{3}; 4\right)$ D) $\left(2; 3\frac{1}{3}\right)$

**17.** Radiusi $\frac{25}{4}$ bo‘lgan sferaga balandligi 8 ga teng bo‘lgan konus ichki chizilgan. Konusning hajmini toping.

A) $96\pi$ B) $72\pi$ C) $144\pi$ D) $192\pi$

**18.** $\frac{3}{\sqrt[3]{9x^2-1}+3} + \frac{2}{\sqrt[3]{9x^2-1}+2} = 2$ tenglama nechta haqiqiy ildizga ega?

A) *4* B) *2* C) *1* D) *0*

**19.** Dastlabki $n$ ta hadining yig‘indisi $S_n = 6n - 2n^2$ formula bo‘yicha hisoblanadigan arifmetik progressiyaning ayirmasini toping.

A) *4* B) $-2$ C) *2* D) $-4$

**20.** $\bar{a}\,(-1;2)$, $\bar{b}\,(-2;1)$ va $\bar{c}\,(-3;2)$ vektorlar berilgan. $k$ ning qanday qiymatida $2\bar{a} - k\bar{b}$ vektor $\bar{c}$ vektorga perpendikulyar bo‘ladi?

A) $-\frac{4}{7}$ B) $\frac{7}{4}$ C) $-\frac{7}{4}$ D) $\frac{4}{7}$

**21.** $f(x) = x^2 + bx + 3$ funksiyaning nollaridan biri 1 bo‘lsa, $b$ ni toping.

A) *3* B) *4* C) $-3$ D) $-4$

**22.** $(x_0; y_0)$ nuqta $y = 3x^2 - bx + 12$ parabola uchining koordinatalari bo‘lsa, $y_0 + 3x_0^2$ ning qiymatini toping.

A) *18* B) *9* C) *15* D) *12*

**23.** Agar $f(x) = (3x-2)^{18}$ bo‘lsa, $f'(1)$ ni hisoblang.

A) *54* B) *18* C) *27* D) *36*

**24.** $n$ natural sonning qanday qiymatida $2 + \frac{1}{1+\frac{2}{n}} = \frac{13}{5}$ tenglik o‘rinli bo‘ladi?

A) *3* B) *1* C) *4* D) *2*

**25.** $\left(\sqrt[4]{2}\right)^{4x+5} = \left(\sqrt{2}\right)^{\frac{-2x}{3}}$ tenglamani yeching.

A) $-\frac{15}{16}$ B) $-\frac{17}{16}$ C) $-\frac{9}{16}$ D) $-\frac{21}{16}$

**26.** $\sin 3x = \cos(x - \frac{\pi}{6})$ tenglamaning $[0;\,\pi]$ kesmadagi barcha yechimlari yig‘indisini toping.

A) $\frac{\pi}{6}$ B) $\frac{\pi}{3}$ C) $\frac{2\pi}{3}$ D) $\frac{5\pi}{6}$

**27.** $\frac{1+mn^{-1}}{(mn)^{-1}} \cdot \frac{m^{-1}}{m^{-1}n - n^{-1}m} : \frac{mn^{-1}}{n-m} \cdot \left(\frac{n}{m}\right)^{-2}$ ifodaning $m = 4$ va $n = 2,5$ bo‘lgandagi qiymatini toping.

A) *1,6* B) *16* C) *10* D) $-1,5$

**28.** Ifodani soddalashtiring: $\left(4a^2b^3\right)^3 : \left(-2a^2b^3\right)^2$

A) $-16a^2b^3$ B) $3a^2b^3$ C) $16a^2b^3$
D) $-3a^2b^3$

**29.** Agar qisqarmaydigan kasrning surati 3 ga orttirilsa, kasrning qiymati $\frac{6}{7}$ ga, maxraji 2 ga kamaytirilsa, kasrning qiymati $\frac{3}{4}$ ga teng bo‘ladi. Berilgan kasrning $\frac{7}{27}$ qismini toping.

A) $\frac{1}{6}$ B) $\frac{1}{12}$ C) $\frac{3}{4}$ D) $\frac{2}{3}$

**30.** Agar $y = \ln(5x+1)^2 - \ln(2x+1)^5 + 4$ funksiyaning grafigiga $(x_0; y_0)$ nuqtada o‘tkazilgan urinma $Ox$ o‘qiga parallel bo‘lsa, $\sqrt{x_0^2 + y_0^2}$ ni toping.

A) *4* B) *5* C) *3* D) *6*

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**Test varianti "Test topshiriqlari to'plami 2019" asosida shakllantirilgan:**

| | | | | |
|---|---|---|---|---|
| 1 – 119-bet 2 | 7 – 36-bet 18 | 13 – 86-bet 12 | 19 – 34-bet 38 | 25 – 55-bet 30 |
| 2 – 68-bet 20 | 8 – 108-bet 15 | 14 – 124-bet 14 | 20 – 114-bet 2 | 26 – 42-bet 79 |
| 3 – 24-bet 21 | 9 – 15-bet 55 | 15 – 28-bet 58 | 21 – 72-bet 30 | 27 – 51-bet 55 |
| 4 – 91-bet 9 | 10 – 9-bet 29 | 16 – 57-bet 17 | 22 – 73-bet 41 | 28 – 46-bet 15 |
| 5 – 19-bet 32 | 11 – 101-bet 82 | 17 – 114-bet 61 | 23 – 80-bet 35 | 29 – 22-bet 51 |
| 6 – 98-bet 60 | 12 – 63-bet 36 | 18 – 66-bet 73 | 24 – 11-bet 18 | 30 – 84-bet 58 |

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